SCIENCE DEPARTMENT • SCIENCE LITERACY & LESSON INTERNALIZATION

# PD AGENDA

# Strengthening Science Literacy: Calibrating Writing Benchmarks, Feedback, and Revision

*Strategy Focus: Teaching Students to Analyze Science Sources With CUBS + CER — adapt a student-facing path from source analysis to an evidence-based explanation, feedback, and revision.*

**WEDNESDAY, OCTOBER 28, 2026 • 2:30–3:30 P.M.** **60 MINUTES** **GRADES 9–12 SCIENCE • BRING FORMS & EVIDENCE**

**TODAY'S OUTCOME** Leave with a ready-to-teach CUBS/CER routine, writing-sample feedback and revision move, Lesson Internalization reflection, colleague strategy, and Friday evidence plan for an authentic course task.

**WHY THIS PLC MATTERS** Every science course asks students to interpret information and explain scientific thinking. CUBS and color-coded CER make source analysis, evidence use, and reasoning visible while students remain responsible for the thinking.

**PROTOCOL CONNECTION**
Every Day Questions 1, 3, and 4; Team Deep Dive Steps 2–4: clarify the learning goal and mastery bar, anticipate student responses and access needs, and plan the student experience, feedback, and revision moves.

**WHAT TO BRING**
One course-specific source or response task; 1–2 de-identified writing samples; current standard(s); student-facing materials; CER-color highlighters; Lesson Internalization Tool; and planning materials.

**FOCUS TO PROTECT**
The Biology benchmark is available only as an optional model. Every science teacher adapts CUBS and color-coded CER to an authentic course source and task. The routine supports students in doing the task; it is not an additional grading system.

## TODAY'S PLC PLAN

| TIME (MIN) | FOCUS | WHAT TO DO | PRODUCT / CHECK |
|---|---|---|---|
| 2:30–2:41 | **Frame the science literacy cycle** | Connect Monday's task experience and Tuesday's planning to the course-wide process: analyze a source, construct an evidence-based explanation, use feedback, and revise. | **Purpose + learning arc** |
| 2:41–3:00 | **Model CUBS + practice a course source** | Think aloud CUBS moves, then apply them to an authentic article, data display, diagram, model, investigation result, or task. The Biology benchmark is optional only. | **Source analysis** |
| 3:00–3:15 | **Color-code CER + plan feedback/revision** | Use a de-identified writing sample to make claim, evidence, reasoning, and linking language visible; plan a prompt and revision action tied to the mastery bar. | **Response + feedback** |
| 3:15–3:22 | **Teacher science literacy strategy share** | Two or three teachers share a concrete CUBS or color-coded CER student routine, access support, feedback move, or implementation insight that colleagues can adapt. | **Colleague learning** |
| 3:22–3:28 | **Course-alike adaptation + LIT reflection** | Use the reflection sheet during PD to record the LIT-aligned task, anticipated thinking, writing-sample evidence, feedback/revision, colleague strategy, and next step; complete the final reflection after PD. | **Ready-to-teach plan** |
| 3:28–3:30 | **Close + evidence plan** | Teach the routine, complete the after-PD reflection and next step, record the evidence to collect, and bring a small de-identified set of completed writing and feedback/revision evidence Friday. | **Friday bridge** |

## PROTECTED INDIVIDUAL WORK WITH COLLEAGUE SUPPORT

**WORK WITH COLLEAGUES**

**WHAT TO DO** Listen to 2–3 strategy shares. Then use the reflection sheet during and after PD to connect the source/task, mastery bar, anticipated thinking, sample-based feedback, student revision, colleague idea, and evidence check.

1. **Select your source + task:** Use a text, graph, data display, diagram, model, investigation result, or scenario.
2. **Model CUBS:** Make source-analysis moves visible before students explain.
3. **Calibrate CER colors:** Help students see the claim, evidence, reasoning, and transitions in a response.
4. **Plan feedback + revision:** Write ready-to-use directions, a feedback prompt, and a revision action.
5. **Share, reflect + collect:** Learn from colleague strategies, complete the reflection, and use evidence Friday to plan a mastery move.

## USE THIS REFERENCE DURING GROUP WORK

**ADAPT THE ROUTINE TO YOUR SCIENCE COURSE**
Begin with an authentic source and response task. Clarify what students must notice, explain, and make visible in a complete response.

**CUBS + COLOR-CODED CER MAKE THINKING VISIBLE**
Students use CUBS to analyze the source and label their claim, evidence, and reasoning so they can see what is present, partial, missing, or not connected.

**FEEDBACK MUST LEAD TO REVISION**
Model the routine, prompt student analysis, and use a criterion-based feedback prompt that leads to a specific revision. Do not complete the science thinking.

**COLLABORATION IN PRACTICE** Protect time for thoughtful individual work; use respectful, student-centered feedback to make the next decision clearer.
**TEACHER RUBRIC CONNECTION** **Standard I: I-A-4 Well-Structured Lessons • Standard II: II-A-2 Student Engagement • Standard II: II-D-2 High Expectations • Standard IV: IV-C-1 Professional Collaboration**

